ALBUKHARY INTERNATIONAL UNIVERSITY

**SCHOOL OF COMPUTING AND INFORMATICS**

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# CCC2143 Software Engineering

**Group Project**

**"BLOOD BANK MANAGEMENT SYSTEM"**

**Prepare for:**

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# 1. Introduction

## I. Project Description

The Blood Bank Management System (BBMS) is a transformative web-based initiative aimed at enhancing healthcare services by optimizing blood donation, storage, and distribution processes. It serves as a vital communication platform connecting blood donors, seekers, and blood banks, with the primary goal of saving human lives. The system integrates real-time GPS functionality, enabling users to locate nearby blood banks and volunteer donors during emergencies efficiently. A centralized database ensures transparent management by maintaining detailed records of blood availability, donor registrations, and recipient information. The BBMS also prioritizes safety through rigorous blood testing and quality control processes, ensuring that all blood stored and distributed meets the highest standards. By aligning with key United Nations Sustainable Development Goals (SDGs), the BBMS plays a significant role in promoting global health equity. It contributes to SDG 3: Good Health and Well-being by facilitating universal access to safe blood supplies and minimizing health risks through stringent quality assurance protocols. Overall, the BBMS addresses critical challenges in blood donation, such as recruiting safe donors and ensuring the availability of adequate blood supplies. By building resilient healthcare systems and promoting sustainable health outcomes, this project makes a substantial contribution to the achievement of the Sustainable Development Goals, particularly in advancing global health and well-being.

## II. Problem Statement

The lack of an efficient communication system between blood seekers, donors, and blood banks often causes delays in securing life-saving blood during emergencies. Existing systems face challenges with real-time availability, donor accessibility, and accurate blood type matching. The process of connecting donors with local blood banks and hospitals is inefficient, leading to delays in blood delivery and compromising the reliability of the supply. This situation highlights the need for a centralized system to streamline interactions between all parties involved. Such a system would ensure quicker responses, better resource management, and improved outcomes for individuals in urgent need of blood.

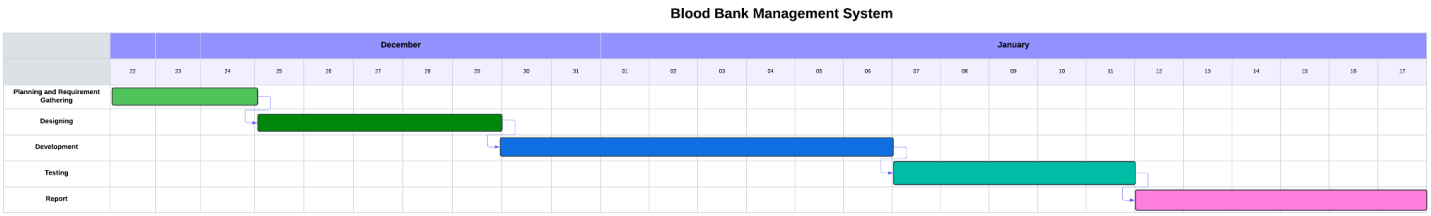
### III. Project Objectives

The Blood Bank Management System aims to create an intuitive platform that facilitates seamless interaction between blood seekers, donors, and blood banks. With simple navigation and efficient tools for registration and blood requests, it ensures easy access to essential services while streamlining communication and coordination among users.

The system incorporates GPS technology to help users quickly locate nearby blood banks, hospitals, and volunteer donors. This feature reduces response times during emergencies, providing real-time location updates to ensure timely access to life-saving resources.

The platform enables real-time tracking of blood inventory and donor registrations, ensuring accurate and updated information. Blood banks can efficiently manage supplies, address shortages, and coordinate with donors, improving the overall effectiveness of blood donation services.

### 2. System Project Planning - Gantt Chart




The following contents are some justifications of the Gantt Chart.

### 1. Planning and Requirement Gathering (Dec 22–24)

Justification: This initial phase is critical to identify and document the needs of all stakeholders, such as donors, hospitals, and administrators. It ensures that the project scope, objectives, and constraints are well-defined, reducing ambiguity in later phases. Deliverables include a requirements specification document that guides the rest of the project.

**2. Designing (Dec 25–29)**

Justification: The design phase transforms the gathered requirements into detailed plans and models. This includes system architecture, database design, and interface mockups. A well-thought-out design ensures the system is scalable, user-friendly, and efficient. It acts as a blueprint for the development phase.

**3. Development (Dec 30–Jan 6)**

Justification: The actual coding and implementation of the system occur here. Developers follow the design specifications to create modules like user management, blood inventory, donor/recipient records, and search functionality. This phase ensures the project's tangible output is built according to plan.

**4. Testing (Jan 7–11)**

Justification: Testing is essential to ensure the system is free of errors and functions as expected. This includes functional, usability, and performance testing. Fixing issues at this stage avoids costly problems during deployment. Deliverables include test reports and a validated system.

**5. Report (Jan 12–17)**

Justification: This phase involves finalizing and documenting the entire project process, challenges, and outcomes. It provides stakeholders with a comprehensive understanding of the project and serves as a reference for future improvements or maintenance.

## 3. System Requirements Analysis

### i. User Requirements

The core functionality of the Online Blood Bank Management System revolves around seamless interaction between its key components: blood seekers, hospitals, blood donors, and blood banks. This system is designed to incorporate all essential elements needed to establish effective communication between these stakeholders, playing a pivotal role in saving human lives. The system reflects the process of identifying, outlining, and organizing the requirements necessary to create an efficient and user-friendly platform.

For blood seekers, the system will provide an intuitive interface enabling them to locate hospitals within their locality using GPS functionality. In emergencies, users can submit blood requests through the platform. Once a request is made, the system ensures the hospital receives it

promptly. The hospital then communicates with the nearest blood bank to check the availability of the requested blood type. If the required blood is available, the hospital places a request with the blood bank to facilitate the transfer. If the desired blood type is unavailable at the first blood bank, the system enables the hospital to automatically search for other nearby blood banks until the need is fulfilled.

From the perspective of blood donors, the system will allow them to identify the nearest blood banks where they can donate blood. Donors can access vital information about the blood banks, such as their location, operating hours, and donation requirements. This feature simplifies the donation process and encourages greater participation in blood donation activities.

**ii. System Requirements: Functional and Non-Functional**

**1. Functional Requirements**

**User Registration and Login**: Users (recipients, donors, hospitals) must register to create an account and log in to access the system.

**Tracking and Identification**: All recipients, donors, and blood types are tracked using unique IDs.

**Donor Search and Donation**: Donors can search for nearby blood banks using their location. Donors can select a blood bank to donate blood.

**Recipient Search and Blood Request**: Recipients can search for nearby hospitals to request blood. The system provides suitable blood matches based on recipient information.

**Hospital Operations**: Hospitals can view the nearest blood banks with available blood stock. Hospitals can request blood from the nearest blood bank and must agree to the blood bank's policies.

**Blood Bank Management**: Blood banks can update bloodstock information periodically.

**Form Submission**: Blood receivers must fill out a request form and agree to the hospital's policies. Hospitals must fill in the required documents and agree to the blood bank's policies before confirming requests.

**2. Non-Functional Requirements**

**Performance**: The system should efficiently handle multiple user searches and transactions without delays.

**Usability**: The user interface should be easy to navigate, providing clear instructions for registration, search, and form submission.

**Scalability**: The system should accommodate increasing numbers of users and transactions, as well as additional data from blood banks and hospitals.

**Security**: The system must securely handle user data, ensuring proper authorization and data encryption. User accounts and sensitive transactions should be protected from unauthorized access.

**iii. Software process model adopted**

The Waterfall model provides a sequential approach to the development of the BBMS, ensuring each phase is thoroughly completed before moving to the next. This approach is ideal for projects where requirements are well understood and stable.

**1. Requirement Analysis:**

Objective: Gather detailed requirements from stakeholders, including blood searchers, donors, hospitals, and blood banks.

Activities: Conduct stakeholder meetings to understand their needs. Document all functional and non-functional requirements.

**2. System Design:**

Objective: Develop the architecture of the system based on the requirements.

Activities: Design database schemas to store user data, blood types, and transaction records.

Create interface design for user-friendly navigation, adaptable to various device sizes.

**3. Implementation (Coding):**

Objective: Develop the system components as per the design specifications.

Activities: Implement user registration and login modules.

Develop modules for blood search, donation, request processing, and stock management.

**4. Integration and Testing:**

Objective: Integrate all system modules and ensure they work together seamlessly.

Activities: Conduct unit tests for individual modules. Perform integration testing to validate interactions between modules.Execute system testing to ensure the system meets all requirements.

**5. Deployment:**

Objective: Deploy the system to a production environment for end-user access.

Activities: Set up the server environment.Deploy the web application and ensure all functionalities are operational.

**6. Maintenance:**

Objective: Provide ongoing support to ensure the system remains functional and up-to-date.

Activities: Monitor system performance and fix any arising issues. Update the system to accommodate any new requirements or improve existing functionalities.

**Advantages of Using the Waterfall Model for BBMS:**

Clear Structure: Each phase has defined deliverables, ensuring clarity in project progress.

Requirement Stability: The model's sequential nature ensures that all requirements are captured and agreed upon before design and development begin.

Easy Management: The linear progression of phases makes project management straightforward, with clear milestones.

**Challenges:**

Inflexibility: Changes in requirements after the initial phases are difficult to implement.
Late Testing: Errors and issues may not be identified until the testing phase, potentially causing delays.

## 4. System Model: use case diagram, activity diagram, sequence diagram and state diagram

### 4.1 Use Case Diagram

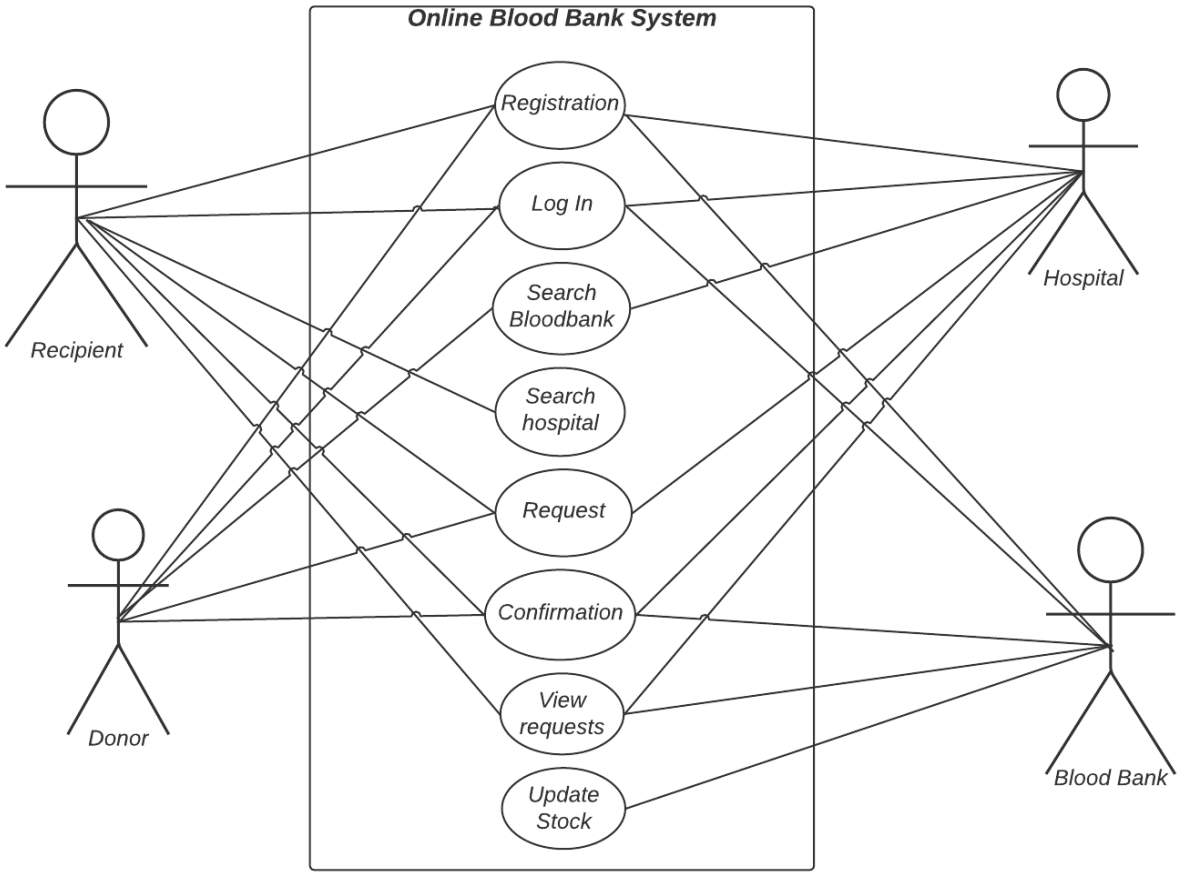



*Figure 4.1 Use Case Diagram*

**Explanation for Use case Diagram.**

The use case diagram above shows that there are 4 actors which are Recipient, Donor, Hospital, and Blood Bank. Every user must register and login in order to use this system. The recipient can search for the nearest hospital and request from the system. The recipient can confirm and view their request. As for the donor, it can search for the nearest blood bank and request from the system. The donor can confirm their request. For the hospital, It can search for the nearest blood bank and request from it. The donor can confirm and view their request. The blood bank can see the confirmation and view requests from other users. Only the blood bank can update the stock from time to time.

**4.2 Activity Diagram**

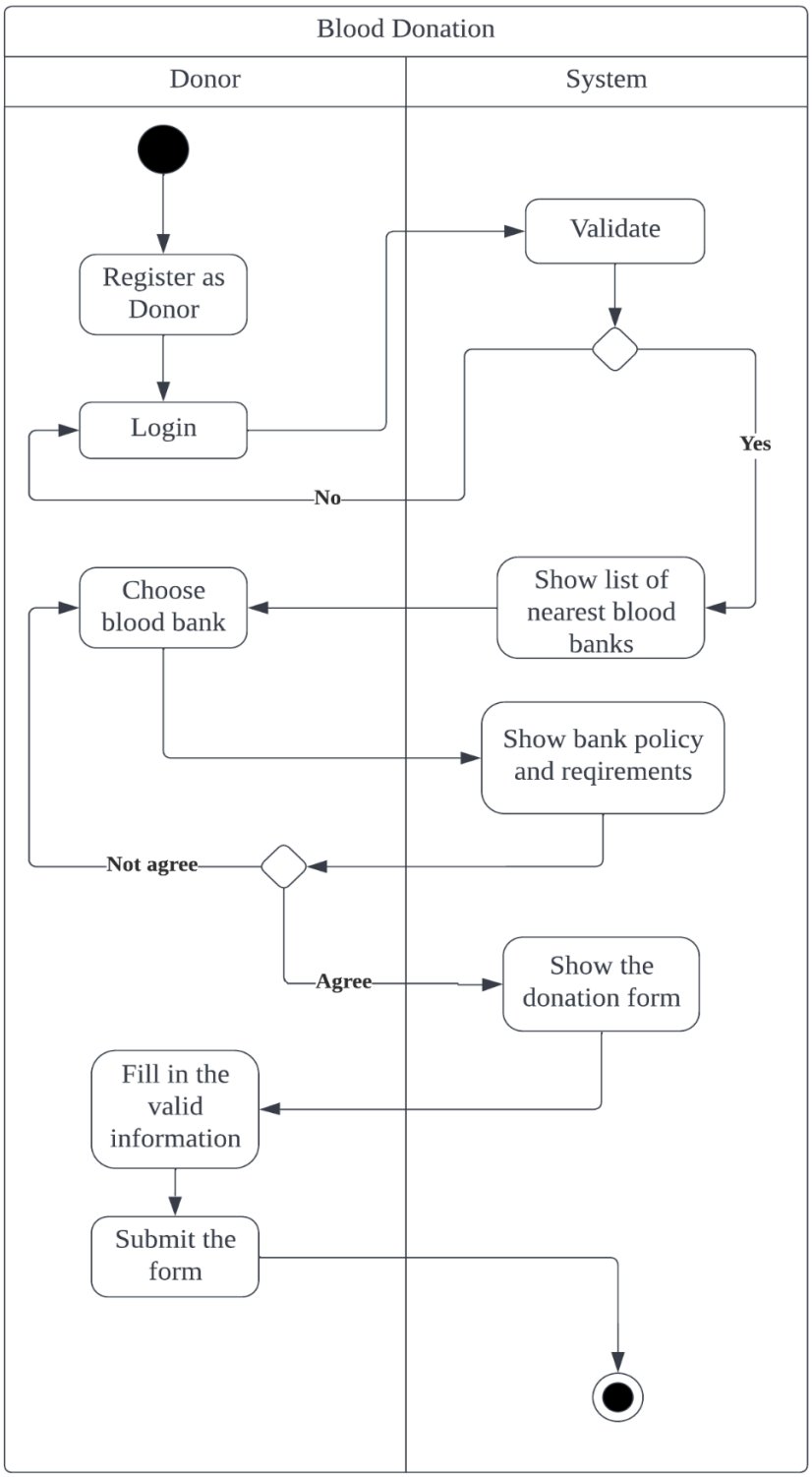



*Figure 4.2.1 Activity Diagram for Donor*

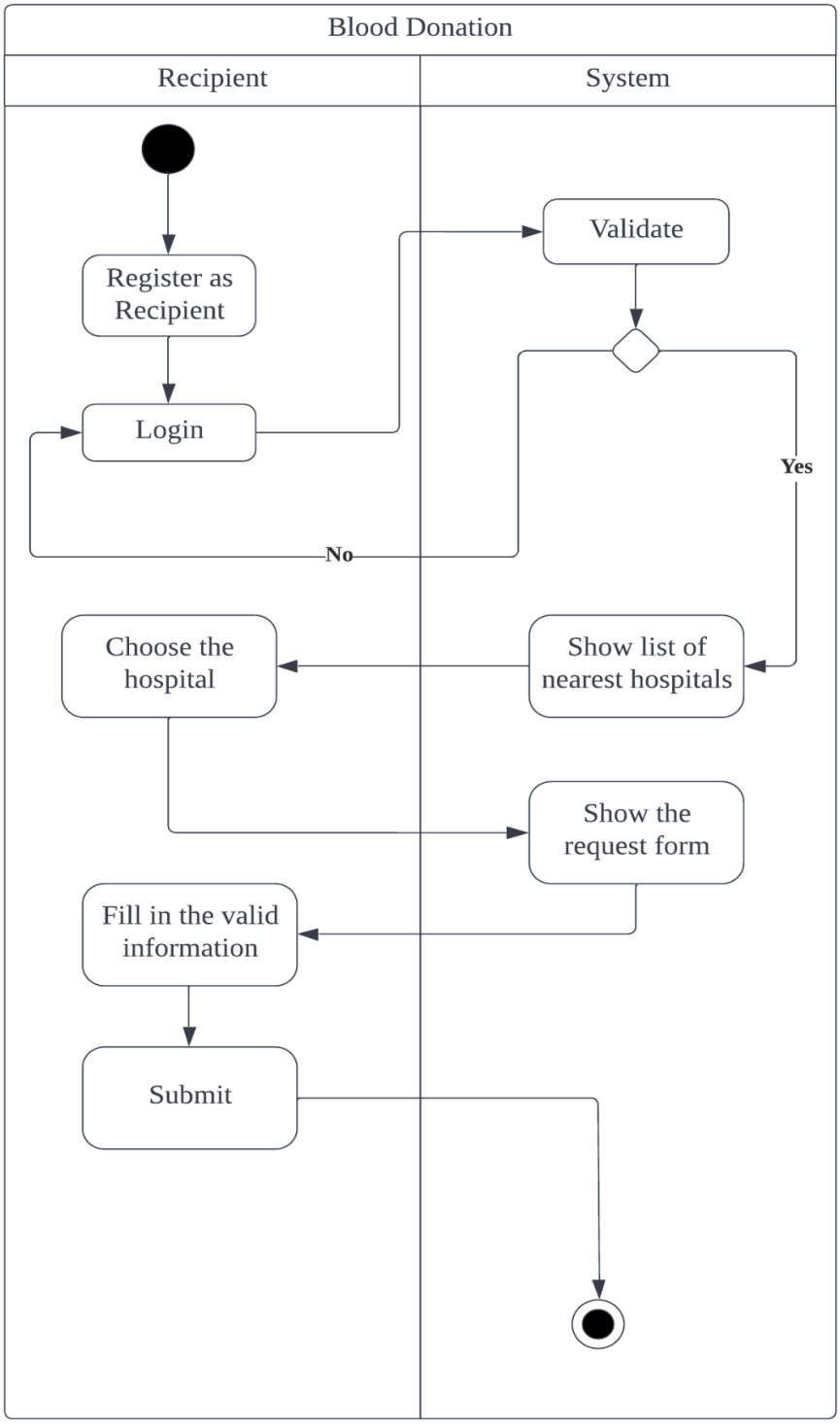



*Figure 4.2.2 Activity Diagram for Recipient*

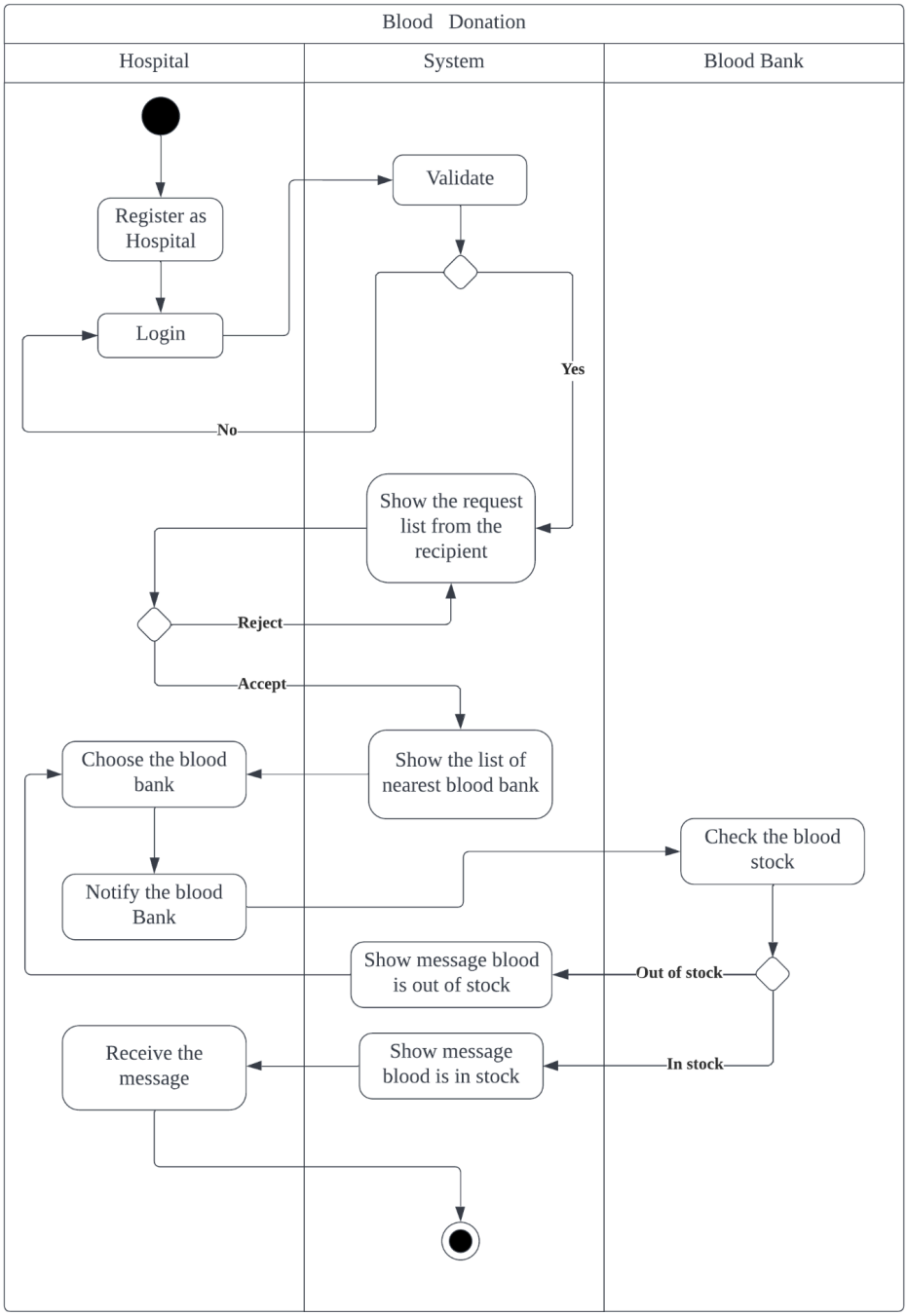



*Figure 4.2.3 Activity Diagram for Hospital*

**Explanation for Activity Diagram.**

1. Every user needs to register (if they are first-time users) and log in as a donor/ recipient/ hospital. After that, the system will validate the information given. If it's not successful, the user must log in again. If it's successful, it will move to the next part.
2. For the donor part, it will show the list of nearest blood banks. The donor can choose which blood bank they want, and the system will show the blood bank policy and requirements. If the donor disagrees with the policies, they can go back and choose another blood bank. However, If the user agrees, the system will show the donation form and the donor must fill in the valid information. After confirming the details, the donor can submit the form, and the system will end.
3. As for the recipient part, the system will show the list of the nearest hospitals. Recipients can choose which hospitals they want, and it will show the request form to the recipient. After the recipient fill in the valid information and submit the form, the system will end.
4. Lastly for the hospital, the system will show the request list from the recipient. The user can choose to reject or accept the request. If the recipient decides to reject a recipient, they can choose another recipient from the list. However, if the recipient chooses to accept the request, the system will show the list of the nearest blood banks. The user can choose which blood bank they want, and it will notify the selected blood bank. The blood bank will check whether the wanted blood is in stock or not. If the blood is not in stock, it will show the message blood is out of stock and the user can choose another blood bank. But, if the blood is in stock, it will show the message blood is in stock and the user will receive the message. The system will end.

## 4.3 Sequence Diagram

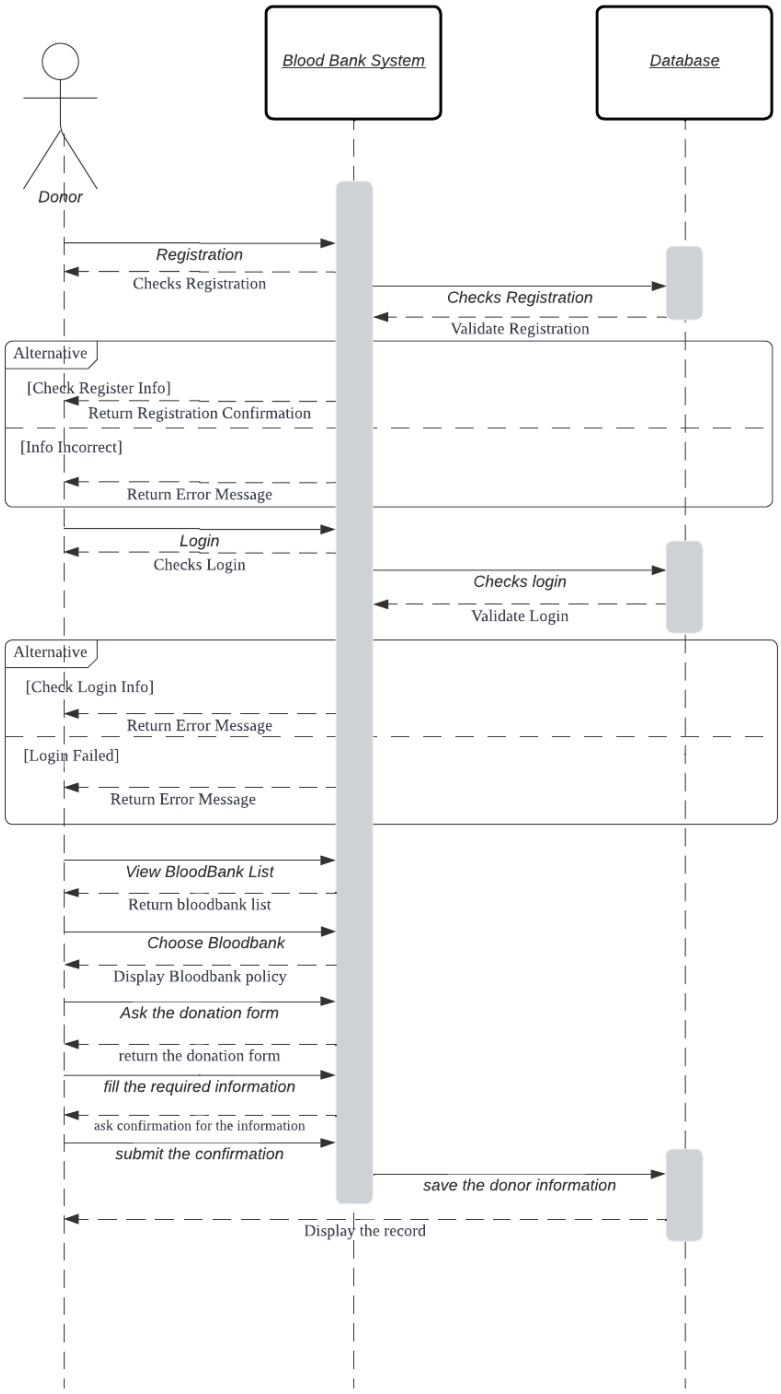



*Figure 4.3.1 Sequence Diagram for Donor*

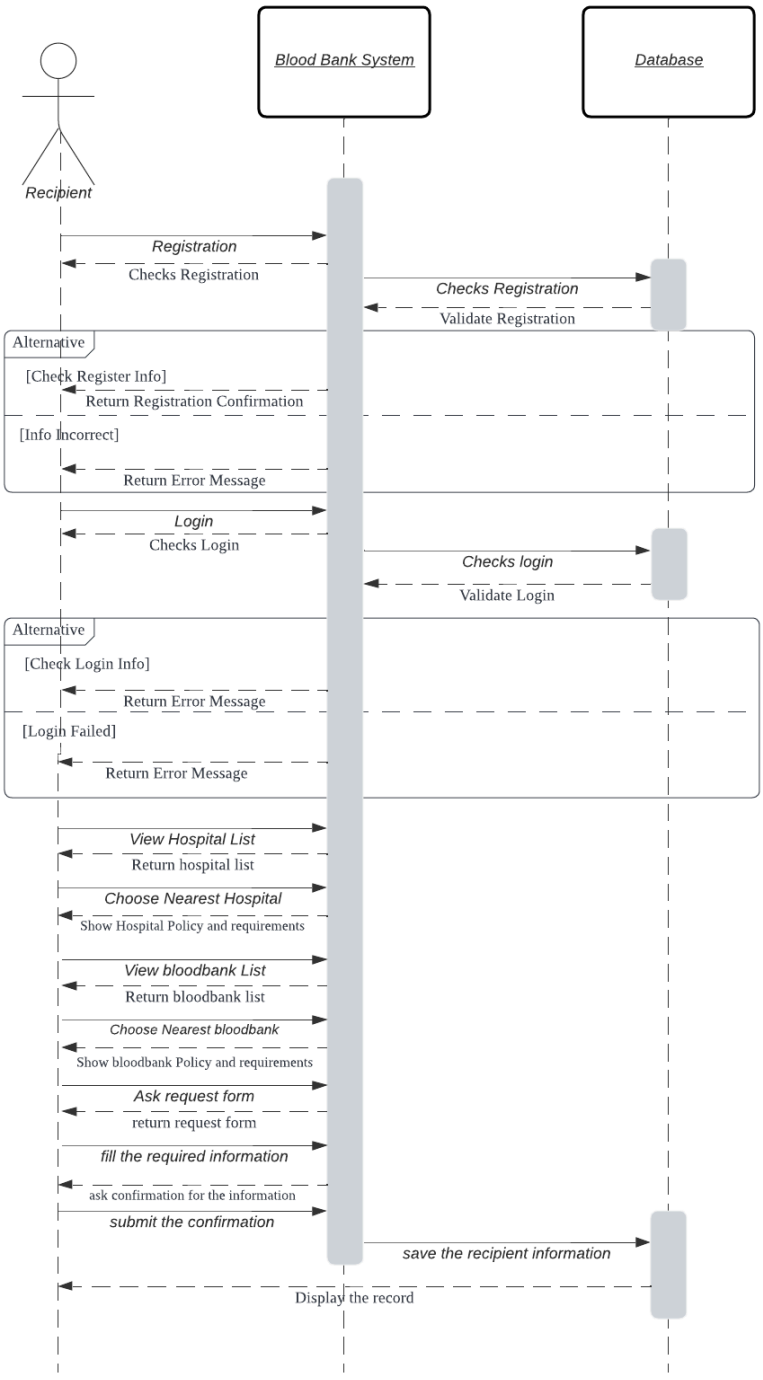



*Figure 4.3.2 Sequence Diagram for Recipient*

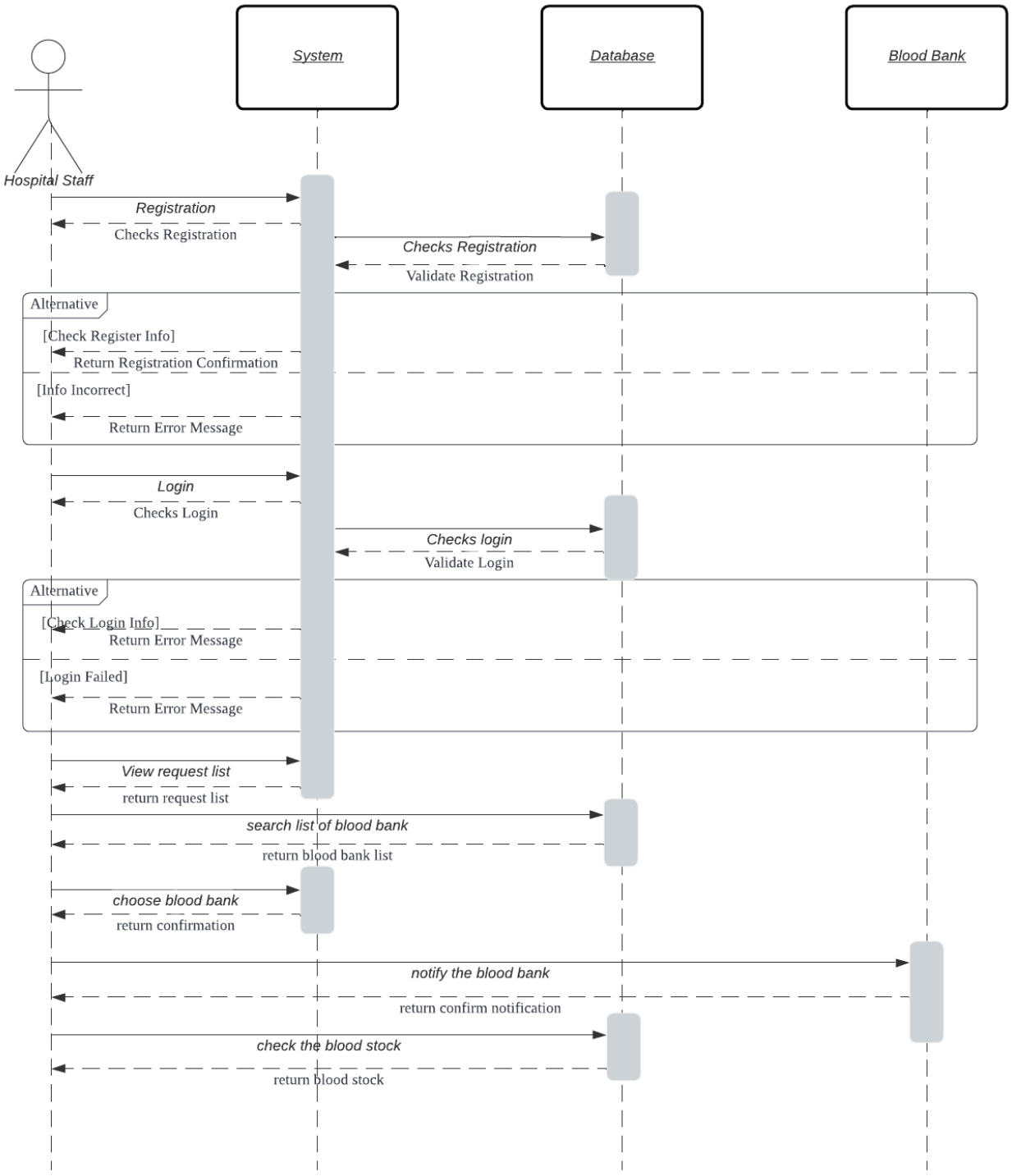



*Figure 4.3.3 Sequence Diagram for Hospital*

**Explanation of Sequence Diagram**

1. The system has three types of end users who have access to the system which are Donor, Recipient, and Hospital Staff. Every user who is first time using the system may need to do a registration first before logging into the system. While registering the profile, the user needs to choose the user type to access the system in a right authenticated manner. Once registration is done, the user needs to login by filling in the correct information to the system. After finishing the filling login information, the system will validate this information with the database to give access to the system or not. If the login information is incorrect, the system will display an error message.
2. On the donor side, the user can view the blood bank list, choose the nearest blood bank, and ask for the donation form to donate blood by understanding the respective blood bank's policy and requirements. After submitting the information to the system, the system will display the record filled by the donor.
3. The user also needs to register and login and the system will validate with the database as mentioned above. The recipient can view the hospital and blood bank lists, and can choose the nearest hospital and blood bank to receive the blood for themselves. And the same with the donor side, the recipient can also ask for the request form to receive blood from the donor by filling in the correct information and blood type to the system. Finally, the system will save the information and display the full record of a certain recipient.
4. The hospital staff can view the request lists from the system, and search blood bank lists that are accepting donations. The staff can choose the available blood bank from the list and can notify the blood bank with the donor request form and lastly, the staff can check if there is enough blood stock for the recipient. The hospital staff side is the interaction with the system and blood bank.

## 4.4 State Diagram

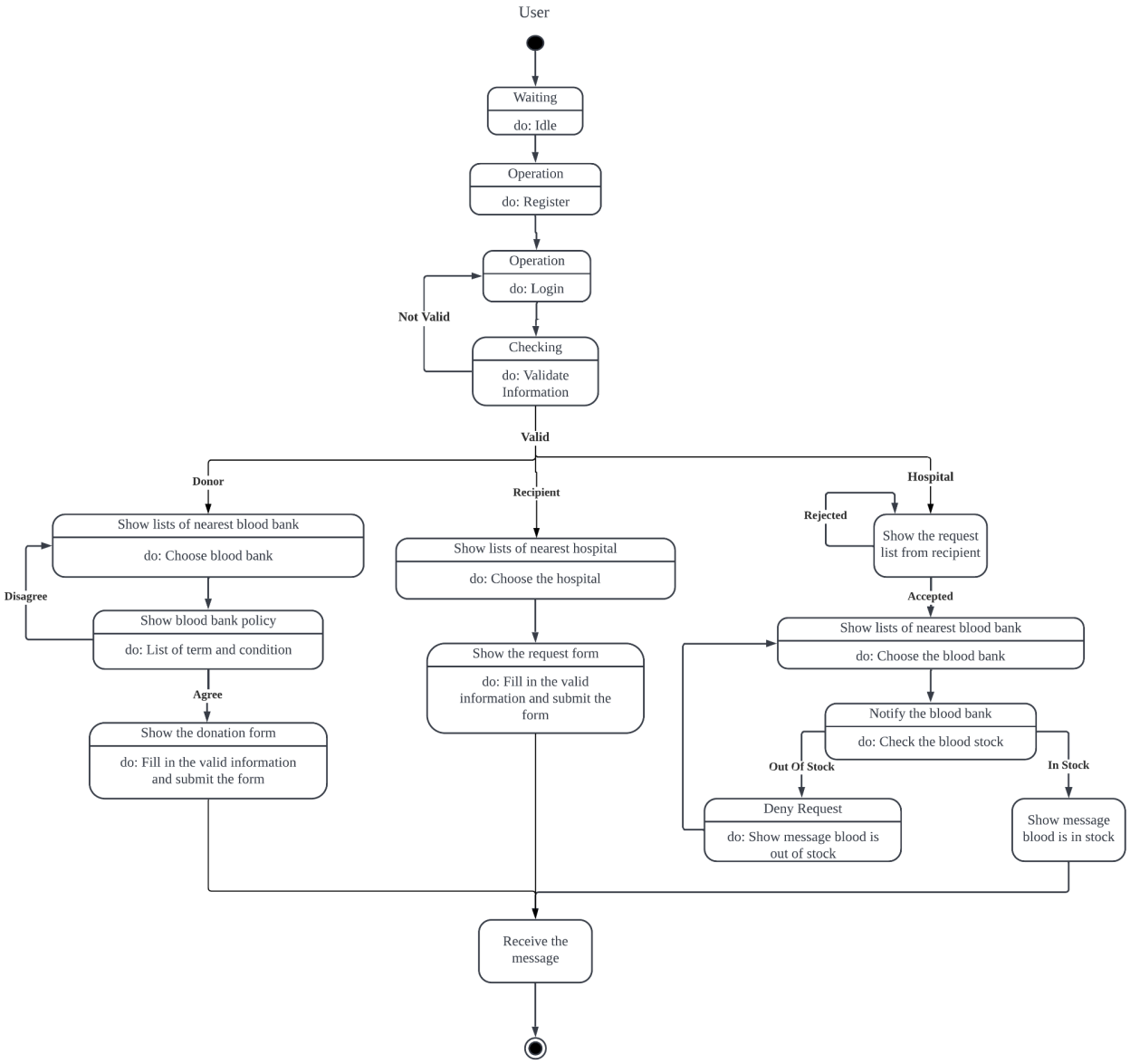



*Figure 4.4 State Diagram*

**Explanation for the state diagram.**

1. First, the user is waiting for the system to load. Every user needs to register (if they are first-time users) and log in as a donor/ recipient/ hospital. After that, the system will check and ask to validate the information given. If it's not valid, the user must log in again. If it's valid, it will move to the next part.
2. For the donor part, the system will show the list of the nearest blood banks and the donor can choose which blood bank they want. After that, the system will show the blood bank policy and requirements. If the donor disagrees with the policies, they can go back and choose another blood bank. However, If the user agrees, the system will show the donation form and the donor must fill in the valid information. After confirming the details, the donor can submit the form, and the system will end.
3. As for the recipient part, the system will show the list of the nearest hospitals, and the Recipients can choose which hospitals they want The system will show the request form to the recipient. After the recipient fills in the valid information and submit the form, the system will end.
4. Lastly for the hospital, the system will show the request list from the recipient and the user can choose to reject or accept the request. If the recipient decides to reject a recipient, they can choose another recipient from the list. However, if the recipient chooses to accept the request, the system will show the list of the nearest blood banks and the user can choose which blood bank they want. It will notify the selected blood bank, and the blood bank will check whether the wanted blood is in stock or not. If the blood is not in stock, it will show the message blood is out of stock and the user can choose another blood bank. But, if the blood is in stock, it will show the message blood is in stock and the user will receive the message. The system will end.

## 5. Implementation of prototype system

The implementation of the prototype system for the Blood Bank Management System (BBMS) involves translating the design specifications into a functional web-based application. The

development phase focuses on creating modules that facilitate user registration, blood donation requests, and inventory management. Key components include:

**Recipient Module:** Recipients can use this module to search for hospitals with available blood supplies, submit blood requests, and receive notifications about the status of their requests.

**Hospital Module**: Hospital staff can manage blood requests, communicate with blood banks, and update inventory statuses. This module ensures efficient coordination between hospitals and blood banks.

**Blood Bank Management Module**: Blood banks can update their inventory, manage donor records, and respond to hospital requests. This module ensures transparency and real-time tracking of blood supplies.The implementation of the prototype system for the Blood Bank Management System (BBMS) involves translating the design specifications into a functional web-based application. The development phase focuses on creating modules that facilitate user registration, blood donation requests, and inventory management. Key components include:

**User Registration and Login Module**: This module allows users (donors, recipients, and hospital staff) to register and log in securely. It ensures that each user type has access to role-specific functionalities.

**Donor Module**: This module enables donors
The prototype system is developed using a combination of front-end and back-end technologies, ensuring a user-friendly interface and robust functionality. The system is designed to be scalable, allowing for future enhancements and integration with additional healthcare services

**6. System Testing Strategy**

The system testing strategy for the BBMS is designed to ensure that the application is free of errors and functions as expected. The testing phase includes the following steps:

**Unit Testing:** Individual modules, such as user registration, donor search, and blood request processing, are tested in isolation to ensure they perform their specific functions correctly.

**Integration Testing**: After unit testing, the modules are integrated, and their interactions are tested to ensure seamless communication and data flow between components.
System Testing: The entire system is tested as a whole to verify that it meets all functional and non-functional requirements. This includes testing the system's performance, usability, and security.

**User Acceptance Testing (UAT)**: The final phase involves testing the system with actual users (donors, recipients, and hospital staff) to ensure it meets their needs and expectations. Feedback from UAT is used to make final adjustments before deployment.
The testing strategy ensures that the BBMS is reliable, secure, and user-friendly, providing a robust platform for managing blood donation and distribution processes

## 7. Challenges and opportunities

### 7.1 Naing Naing ( Project Manager)

In my role as a project manager, I encountered and conquered a lot of challenges and take chances from different aspects. Balancing the demands of the project with my other responsibilities can be difficult. Working with team members who have varying schedules, skills, and levels of commitment can be challenging. While I am managing this project, I may not have extensive experience in handling large-scale projects like this one. This can make it harder to make quick decisions and resolve conflicts. Without real-world project management experience, I faced difficulties addressing unexpected issues due to limited resources or time constraints. However, as the project manager, I have the opportunity to further develop my leadership abilities by guiding my team, delegating tasks, and making critical decisions. Managing this project provides me with hands-on experience in planning, scheduling, and executing a complex project.

### 7.2 Zayar Linn

One of the challenges we faced was understanding how to properly represent the system's processes in the diagrams. Sometimes, our group misunderstood the requirements, which meant we had to go back and fix the diagrams. Keeping all the diagrams consistent as the system design changed was also tricky and took a lot of time. Another challenge was making sure everyone in the group was on the same page when interpreting and working on the diagrams. Despite the challenges, there were many opportunities. Creating these diagrams helped us understand the system better and made it easier to explain how everything works. They also helped us spot problems early and fix them before moving on to development. The diagrams made it clear which parts of the system each group member could work on, so we could divide the work efficiently. Overall, this project helped us learn more about system design and teamwork, which will be useful in the future.

### 7.3 Hashem

One of the biggest challenges I faced was the system's technical complexity, including user registration, blood inventory management, and GPS integration, especially within a tight timeline. Integrating modules, ensuring data security, and designing a user-friendly interface for diverse users required significant effort. Balancing the project with other responsibilities was also tough. Despite these hurdles, I gained valuable skills in web development, database management, and system integration. This experience enhanced my problem-solving, teamwork, and communication skills, and showed me the real-world impact of our work, boosting my confidence for future challenges.

### 7.4 Omid

As a front-end and back-end developer on the Blood Bank Management System (BBMS) project, I faced challenges and opportunities. On the back end, managing the database for blood inventory, user data, and transactions required ensuring consistency, avoiding redundancy, and maintaining efficiency. Handling sensitive user data securely involved encryption and best

practices. Balancing front-end design with back-end functionality demanded careful coordination. Despite these challenges, I enhanced my full-stack development skills, particularly in integrating front-end and back-end systems. I also gained hands-on experience in database security and optimization, which are crucial for career growth.

### 7.5 Ahaed Alam Mehedi

As a front-end developer, I faced challenges in creating a responsive and intuitive interface for diverse users while ensuring seamless integration with back-end functionalities. Balancing design iterations, testing, and tight deadlines was demanding. However, these challenges offered opportunities to enhance my skills in responsive design, user experience optimization, and teamwork. This project strengthened my ability to create user-centered designs and improved my coordination with other team members.

## 8. Roles and responsibilities of project members

### 8.1 Naing Naing

As a project manager for a blood bank management system, my responsibilities include structured team management through clear role allocation and fostering a supportive environment. I help to start and finalize the report. And focused on designing and illustrating the Use Case Diagram of the system also. For the Use Case Diagram, I identified all potential actors and their responsibilities, highlighting the functional features they utilize. This provided a clear overview of the system's capabilities. Moreover, including drawing Gantt Charts, identifying and mitigating potential risks, and planning for contingencies, are key to ensuring the project's timely and efficient completion in alignment with its objectives.

### 8.2 Zayar Linn

My responsibilities in this project focused on designing and illustrating the Activity Diagram, Sequence Diagram, and State Diagram for the system. I analyzed the system workflows and represented them visually, ensuring the diagrams accurately captured the system's processes and transitions. For the Activity Diagram, I detailed the step-by-step actions performed by donors,

recipients, and hospital staff, highlighting decision points and alternate paths. In the Sequence Diagram, I illustrated the interactions between the system components and users, ensuring logical and consistent representation of workflows like blood donation requests and inventory updates. The State Diagram showcased the system's states and transitions, such as user registration, form validation, and blood availability checks.

Throughout this process, I worked to ensure the diagrams were clear and aligned with the project requirements. Although I faced challenges in managing changes and maintaining consistency across diagrams, I collaborated with my teammates to address these issues. Overall, my role contributed significantly to improving our understanding of the system and ensuring its processes were well-documented for development.

### 8.3 Hashem

In this project, my responsibilities were divided between front-end and back-end development. On the front-end, I worked on designing and implementing the user interface, ensuring it was intuitive and easy to navigate for donors, recipients, and hospital staff. This involved creating responsive layouts, integrating GPS functionality for locating nearby blood banks, and ensuring a seamless user experience across different devices. On the back-end, I focused on developing the core functionalities, such as user authentication, blood inventory management, and database integration. I also worked on ensuring secure data handling, implementing encryption for sensitive information, and optimizing the system for performance and scalability. Collaborating with the team, I helped bridge the gap between design and functionality, ensuring the system was both visually appealing and technically robust. While balancing these tasks was challenging, it was a rewarding experience that allowed me to grow my skills in both front-end and back-end development.

### 8.4 Omid

My role for this project included both front-end and back-end development. I created and put into place a user-friendly front-end interface to guarantee seamless interactions between hospital employees, donors, and recipients. My efforts concentrated on developing adaptable layouts, user-friendly navigation, and a unified visual style that improved the user experience on a range

of devices (*Web Forms - Learn Web Development | MDN*, 2024). In order to efficiently assist consumers in locating nearby blood banks, I also included capabilities like GPS tracking. I created and maintained the back-end's essential features, including user registration, authentication, and blood inventory control. By putting database integrations into place, streamlining searches, and encrypting private data, I made sure that data handling was safe and effective.

### 8.5 Ahaed Alam Mehedi

In this project, my primary responsibility was front-end development, where I designed and implemented user-friendly interfaces to ensure seamless interactions for donors, recipients, and hospital staff. I focused on creating responsive layouts, intuitive navigation, and a visually cohesive design to provide an exceptional user experience across various devices. Additionally, I worked on integrating advanced features such as GPS functionality to help users locate nearby blood banks effortlessly. My role was pivotal in bridging the gap between user needs and the system's technical aspects, ensuring a smooth and engaging user interface that aligned with the project’s goals.

## 9. Conclusion

The Blood Bank Management System (BBMS) project achieved its objectives of designing a transformative web-based platform to optimize blood donation, storage, and distribution processes. The system provides an efficient means of communication between donors, recipients, hospitals, and blood banks, addressing critical challenges in blood management, such as delays in response, real-time tracking, and accurate blood type matching.

The project utilized a structured development approach, adopting the Waterfall model, which ensured that all phases, from requirement gathering to testing and implementation, were thoroughly planned and executed. The prototype system integrates critical functionalities, such as GPS-based location tracking, secure data handling, and real-time inventory updates, offering a scalable and user-friendly solution.

By aligning with Sustainable Development Goal 3: Good Health and Well-being, this project contributes to promoting global health equity. Despite challenges such as tight timelines, system

complexity, and maintaining data security, the team overcame them through collaboration and innovation. The BBMS stands as a robust and reliable platform, ready to make a meaningful impact on blood donation and healthcare services.

## 10. References


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**Appendix**

**Here is our drive link for the appendix. link**